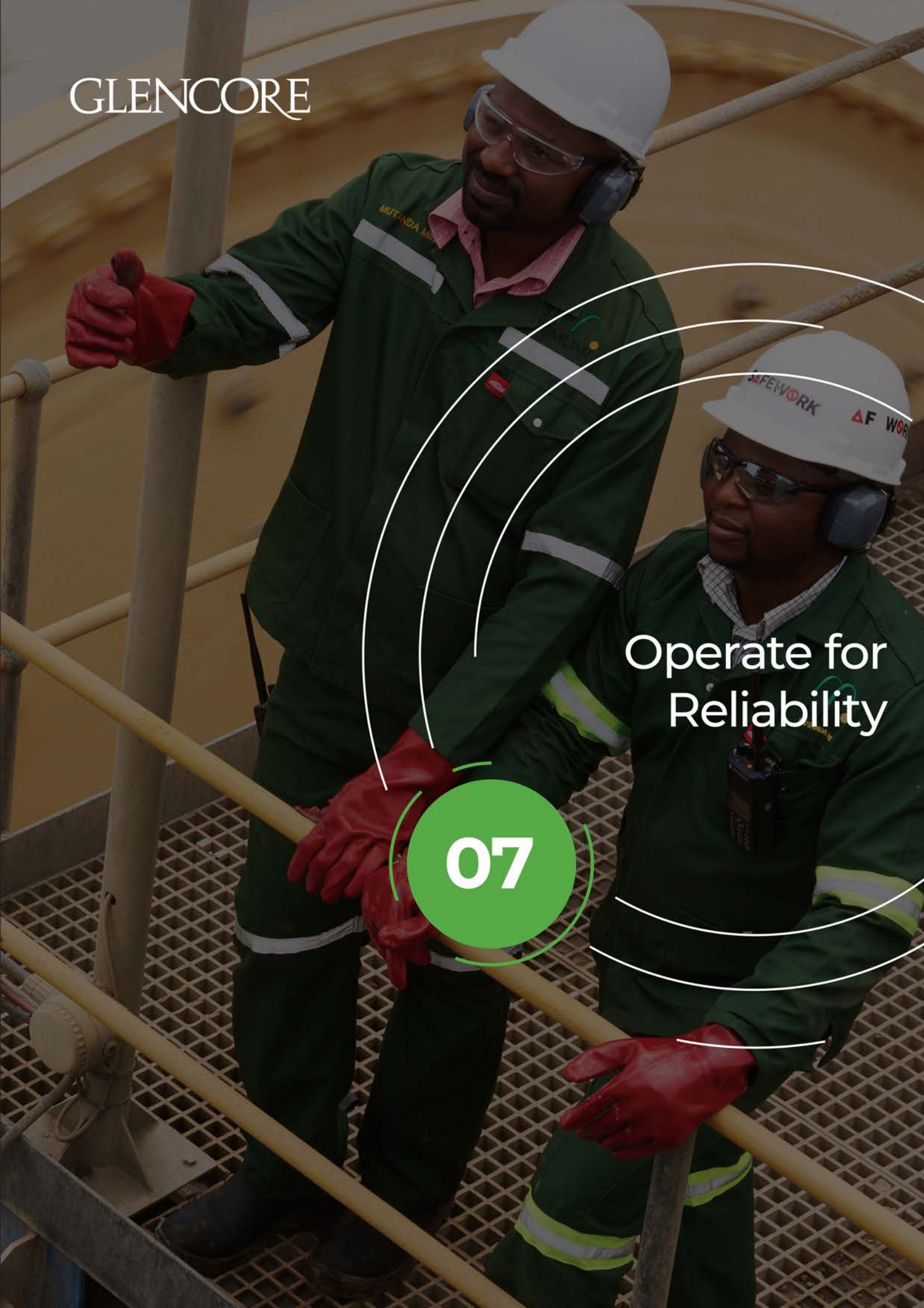
GLENCORE



Operate for Reliability

07

07

Operate for Reliability

Asset Management, Glencore Copper
11 October 2021, Version 1.0

Contributors:

- Maximo Rojas (PASAR)
- Felipe Bunout (Lomas Bayas)
- Ryan Turner (Mutanda Mining)
- Danai Saotayanan (Queensland Metals - Processing and Smelting)
- Tom Barber (Ernest Henry Mining)
- Mark Backman (Queensland Metals – Mining)
- Christopher Dwan (Queensland Metals – Processing and Smelting)
- Neale Laffin (Queensland Metals – Processing and Smelting)
- Felipe Moraga (Technology Transformation and Operational Excellence South America
- Osvaldo Zamorano (Operational Excellence Metallurgical Assets),
- Peter Zuliani (Operational Excellence Metallurgical Assets),
- Dino Moll (Glencore Copper Asset Management - Australia),
- Rohan Reisener (Glencore Copper Asset Management - Australia).

Supported by the AM Network Pty Ltd

Preface

The Asset Management Framework, Policy, and Manuals are intended to provide leading Asset Management practices when applied and will be continually improved for the benefit of Glencore Copper.

The real value of sharing the knowledge gained from the use of these documents will come from continued discussion of the issues and rationale behind each document rather than by reading and interpreting them as individuals.

Readers are encouraged to consider these against our values around Safety, Responsibility, Simplicity, Integrity, Openness and Entrepreneurialism and how they reflect our Purpose – to responsibly source the commodities that advance everyday life. This document lays out the minimum standard for Glencore Copper and compliance to this manual will demonstrate a maturity value equal to Planned/Deployed (3) on the Glencore Copper Asset Management Maturity Scale.

Table of contents

Preface 2
Table of contents 3
Figures 3
Tables 3
1. Introduction 4
2. Objectives 4
3. Definitions 5
4. Approach 7
5. Deployment 17
6. Measures of Success 18
7. Improvement 20
8. References 20
9. Revision History 20
10. Appendix A – TARP Example 21
11. Appendix B -Operator Scorecard Example 22
12. Appendix C – Equipment Score Card Example 23
13. Appendix D – Stakeholder Signoff 24

Figures

Figure 1-1 Asset Management Framework 4
Figure 2 – Operate For Reliability Flow 8

Tables

Table 4-1 Process Description 9
Table 6-1 Measure of Success 18
Table 8-1 References 20

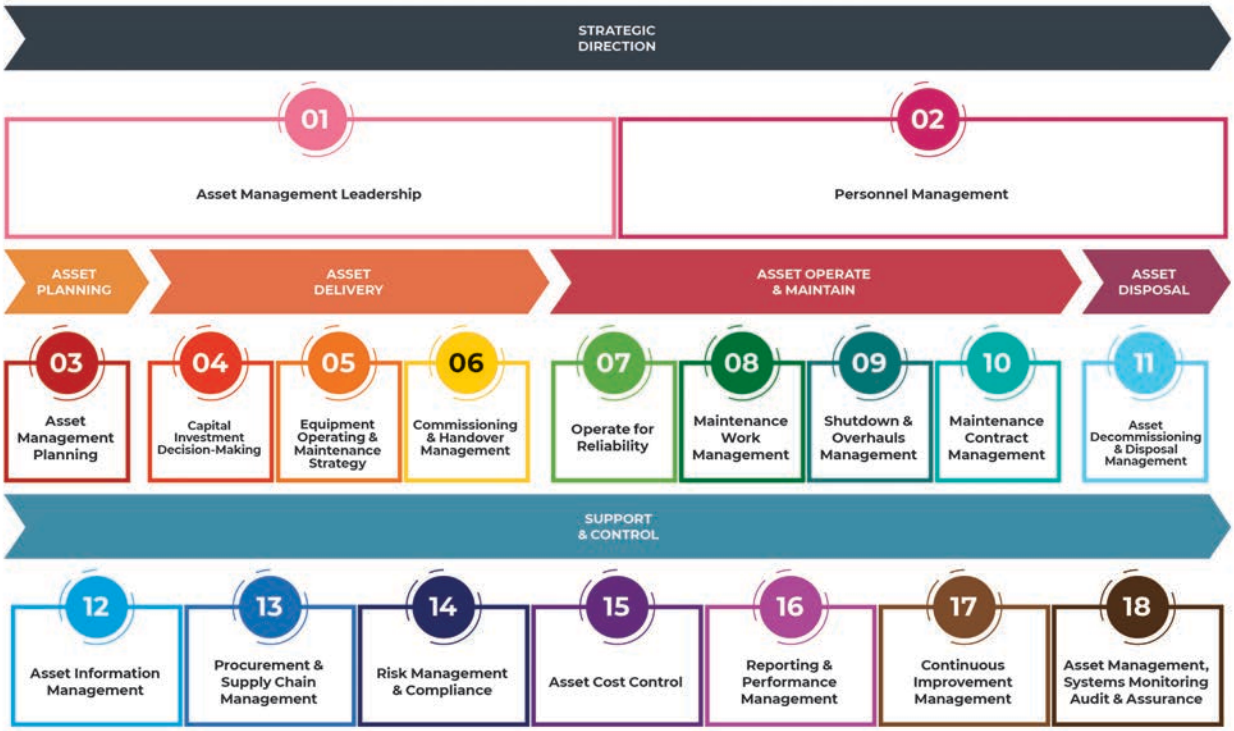
Copyright © Glencore 2022

1. Introduction

The Asset Management Framework, Policy, and Manuals define how Glencore Copper manages the whole of life of an asset to ensure optimal value is extracted by ensuring that assets are capable of meeting business and stakeholder objectives.

Figure 1-1 describes the high-level Asset Management Framework, which enables Glencore Copper to deliver its Asset Management purpose and vision. Each component of the model is described below in further detail.

FIGURE 1-1 ASSET MANAGEMENT FRAMEWORK



2. Objectives

The objective of this manual for Operate for Reliability is to define the approach, deployment, measurement of results, and continuous improvement to ensure we have competent people operating our equipment within design parameters and reporting defects in order to optimise equipment reliability and productivity.

2.1 Accountabilities

Dedicated resources must be assigned, and capabilities developed within each Operation to identify the key aspects required to ensure that equipment is operated to optimise reliability and productivity.

- Assign accountability for each or all of the following:
  - Operate for Reliability identification, planning, scheduling, resourcing, execution and history recording.
  - Operate for Reliability procedures defined and documented.

Copyright © Glencore 2022

- Define and document Operate for Reliability competencies required by Global Job Groups for the operation of Glencore Cu equipment.
- Plan, deliver, and record Operate for Reliability training to support the embedment of required Operate for Reliability system and process competence and knowledge.
- Provide sufficient CMMS user access points to operational staff to enable the daily entry of Maintenance Work Management information into CMMS.
- Roles defined regarding accountability for how systems and assets are operated and maintained to ensure Asset Management objectives are delivered.

2.2 Scope

Operate for Reliability processes must be utilised within Operations for the development and continual improvement for the operation and maintenance of Glencore Cu equipment.

The scope of Operate for Reliability Manual includes:

To monitor equipment operation that is outside design/operational parameters, and provide feedback to accountable and responsible personnel, to facilitate the correction of out of specification operating practices and/or operational conditions to:

- Increase asset life as a result of adoption of best operational practices.
- Reduce unplanned maintenance events.
- Reduce Total Cost of Ownership (TCO)
- Reduce organizational risk exposure.
- Improve safety outcomes and improve operational performance.

This manual should be read in conjunction with the requirements of Element 05 Equipment Operating and Maintenance Strategy to ensure that maintenance tactics are current and effective.

3. Definitions

Operate for Reliability.

- Assist operators to understand how their operating practices and techniques can impact the reliability of equipment.
- Demonstrate correct practices and techniques for particular operating applications.
- Assist in embedding leading operating practices.
- Be used in conjunction with the training process to improve operator's skills.
- Provide the tools for supervisors/managers/trainers to facilitate to points above.

Reliability - Reliability is defined as the probability that a component (or an entire system) will perform its function for a specified period of time, when operating in its design environment

Operator Routines – Work undertaken by operations personnel to ensure that the operating equipment is performing as expected and to identify issues and to create work requests to rectify issues.

Equipment Health –. The physical condition and performance capability of operating equipment.

Failure Mode Effect and Criticality Analysis (FMEA) - is a structured approach to discovering potential failures that may exist within a product, asset, or process. Failure modes are the ways in which a process can fail. Effects are the

Copyright © Glencore 2022

ways that these failures can lead to waste, defects, or harm. Failure Mode and Effects Analysis is designed to identify, prioritize, and limit these failure modes.

Asset Criticality – a process that enables the ranking of systems assets in order of business importance to ensure that effort applied is directionally proportionate to the criticality and impact to the business. Criticality is relevant for the business decision making process to manage business risk and value. Criticality is typically used to determine:

- As an input to determine the overall priority for performing a maintenance task (sometimes combined with a "Work Order Priority" entered against the specific task to give an overall priority for the task)
- To determine, at a high level, the type of risk mitigation strategy to be applied to the equipment (e.g. do condition monitoring and defect elimination on high criticality items)
- As an input into determining the optimum spare parts holdings required for the equipment item
- To provide input into the capital program so that "high criticality" equipment is given a higher priority for upgrade or replacement.
- To guide reliability engineers so that they focus their reliability improvement efforts on the most "critical" equipment.

Operating Context - The current condition, environment, and culture in which you operate a piece of equipment. This would include but not be limited to the following:

- Temperature (Hot, Cold or severe swings); Dirty or dusty atmosphere; Wet or dry area; Corrosive, erosive or abrasive environment; Dark or dimly lit; Noisy; Culture (Goals and expectations not clearly defined, high level of emergency/demand work); Operating outside design expectations or performance standards; Asset condition (Loose, improperly supported, improperly installed, improper design, damaged); Improper operation (Start up, shutdown, product change, setting, speed, flow, pressure); Human error (I forgot, no check lists or procedures)

Reliability 101 – The basics of Reliability Engineering. Understanding the behaviour of equipment over its lifetime will provide insight as to why proactive maintenance (as opposed to repairs) is critical.

Interface - To merge or mingle, bonding and synthesizing by communicating and working together.

Operational Variance Analysis and Response Plans - Defines the minimum set of actions required by operators in response to a variation from normal operating conditions.

5S – Is a Lean Manufacturing approach and consists of 5 key steps for improvement:

- Sort
- Set in Order.
- Shine
- Standardise
- Sustain

Critical System – a productive system comprising of maintainable assets (critical equipment) that if unavailable would have a significant risk, cost and production impact to Glencore Cu.

Copyright © Glencore 2022

4. Approach

The approach for Operate for Reliability is described through:

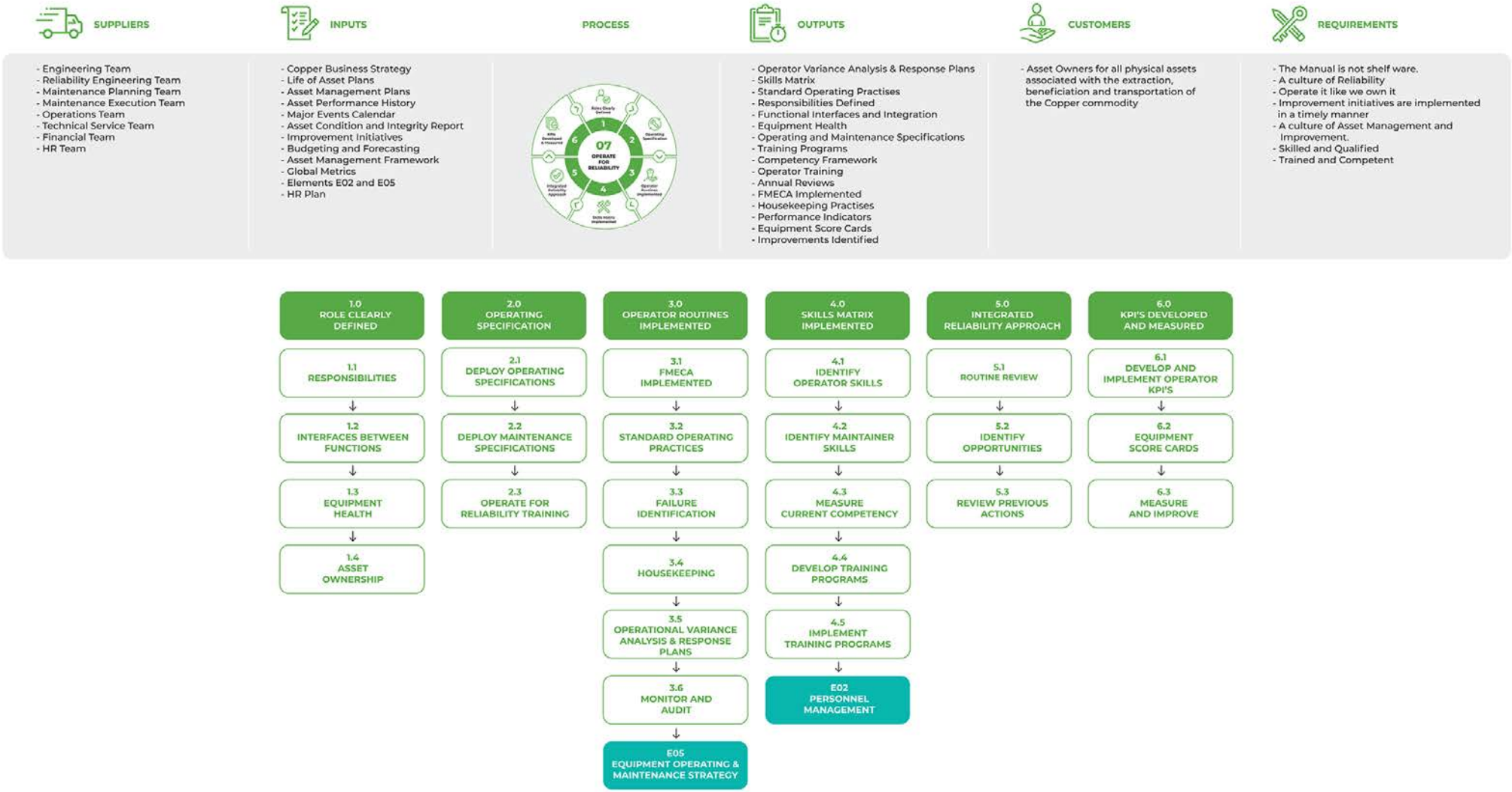
- Process Flow - pictorially illustrates the key Operate for Reliability activities, refer to Figure 2. The intent of the workflow is to ensure that equipment is operated withing design specifications and to understand that Reliability is the responsibility of Operations, Maintenance, Engineering and Supply functions. While the workflow does not provide an end-to-end process map, it does provide the key process areas that have been considered and included in this manual and as detailed in the Glencore Cu Asset Management Framework document.
- Process Description – defines in more detail the key Operate for Reliability activities, refer to Section 4.2.
- Process RACI Matrix - indicates for each activity, the positions who are Responsible, Accountable, who are Consulted prior to final decision/action, and those who are Informed of any outcomes and actions, refer to Section 4.3

Copyright © Glencore 2022

Operate for Reliability

4.1 Process Flow

FIGURE 2 – OPERATE FOR RELIABILITY FLOW



 Copyright © Glencore 2022

4.2 Process Description

TABLE 4-1 PROCESS DESCRIPTION

PROCESS STEP	DESCRIPTION
1.0 Roles Clearly Defined	**Defining clearly and regularly communicating the roles and responsibilities of the technical teams, operators and maintainers**
1.1 Responsibilities	Reliability is everyone's responsibility. Equipment operators generally have a closer connection with the equipment than the maintainers. Building a collaborative partnership between production and maintenance, in which production owns the operation of the equipment and maintenance and reliability provides sufficient capacity to ensure that equipment continues to function as expected. This creates the optimal balance between operations, production and reliability functions. Responsibilities shall be fully developed and documented. There shall be clear lines of communication set up within and between functional departments. Responsibilities shall focus on those activities and actions aimed at ensuring optimal equipment reliability for which everyone has a responsibly to contribute.
1.2 Interfaces Between Functions	Identify the System / Asset that require Equipment Operating and Maintenance Strategy. This can come from business needs such as risk reviews, incident investigations, asset acquisition, modification, and data analysis. The impacts of Equipment Operating and Maintenance Strategy on Systems and Asset shall be fully understood.
1.3 Equipment Health	Understanding equipment health can yield significant reliability improvement and therefore it is necessary that those who are intimate with the operational equipment escalate any unusual operating parameters such as condition or operating limit including noise, heat, vibration, smell, decreased output and the like. A notification shall be entered into the CMMS detailing the issue regarding asset health and Element E08 Maintenance Work Management shall be followed to ensure corrective work is performed and the asset returned to service. Where available, monitoring systems can be used for identification purposes, such as: • SCADA • Fleet Management • Alarm system Equipment Health shall be monitored and reported, each site shall determine reporting format and timeframes.
1.4 Asset Ownership	A question that is often asked "Who owns the assets". The short answer is, everyone that has a vested interest in how assets are purchased, operated, maintained, and disposed of. Asset ownership is a joint responsibility and therefore cross functional collaboration is essential for Asset Management success.
2.0 Operating Specifications	**Communicate and adhering to assets' design operating envelope and maintenance parameters by all departments**
2.1 Deploy Operating Specifications	Assets are purchased to provide value to the organisation and are designed to provide reliable and sustainable operations. That is, as long as the assets are operated within design capability and operational windows. Operating assets outside this capability will reduce the asset expected operating life, will be less reliable and be more costly to Glencore Cu. It is also important to recognise that demand requirements can

Copyright © Glencore 2022

	impact on production rates and that increase in production rates could have a detrimental effect on asset health. Production rates should be analysed to ensure that the equipment can continue to operate within their optimised operating window and with consideration of design capability. Should there be a requirement to operate outside design parameters then a cost benefit analysis shall be conducted, and Glencore Cu Change Management process followed. All personnel who are responsible for the operation of asset shall have intimate knowledge of the asset operating specifications and limits and have knowledge to understand when an asset enters in potential failure. This will ensure that corrective work can be planned before asset functional failure occurs. Operating specifications and tolerances shall be identified and documented by the responsible person.
2.2 Deploy Maintenance Specification	Please consult E05 Equipment Operating and Maintenance Strategy Variable demand can have a significant impact on maintenance delivery regarding cost and risk. Should production demand change an adjustment to the maintenance tactics shall be undertaken to adjust the frequencies of each task and other attributes as deemed necessary. From an Operate for Reliability perspective maintenance tactics shall be developed with consideration to: • Design Specification • Regulatory, legal and HSEC requirements • Operating context • Failure frequency with respect to variable demand • Standard Operator inspections Maintenance Supervisors shall ensure that maintenance quality control is established and documented to ensure maintenance rework is eliminated. Reliability Engineers shall ensure that maintenance quality assurance is established and documented to ensure that the maintenance tactics which are deployed are effective.
2.3 Operate for Reliability Training	All operation and maintenance teams shall undertake Operate for Reliability training Each operation shall develop their own training requirements, but could typically include: • Failure Modes, Effect and Criticality Analysis (FMECA) • Equipment Operational and Performance Standards • How operating equipment has an impact on performance. • Understanding failure modes and the impact of those failure • Operational Routines • Housekeeping

Copyright © Glencore 2022

	• Monitoring and Improvement
3.0 Operator Routines Implemented	**Develop, communicate, and implement operator routines.**
3.1 FMECA Implement	Please consult E05 Equipment Operating and Maintenance Strategy The development of maintenance strategies and tactics in covered in Element 05 Equipment Operating and Maintenance Strategy. Operator Routines shall be developed as part of that process and Operations Personnel shall form part of the FMECA team as detailed in E05 manual. This will enable operators to recognise: • Failure modes • Failure effects • Failure consequence • Failure frequency • Failure detectability • Failure risk • Failure prioritisation • Improvement actions • Post action failure risk This will enable: • Early identification and elimination of potential product/process failure modes • Prioritisation of product/process deficiencies • Understand problem prevention. • Understand risk and actions taken to reduce risk. • Increase in equipment uptime and reduce associated cost. • Teamwork and idea exchange between functions • Understanding the business impact of how equipment is operated and maintained
3.2 Standard Operating Practices	Operations to develop and implement standard operating practices to ensure that systems, assets and equipment are operated on a consistence basis, regardless of whether fixed or mobile plant. Operator routines such as shift checks and standard operating tasks that address failure modes or safety requirements are to be set up in the CMMS and E08 Maintenance Work Management followed. Tasks are to be monitored for completion and corrective work identification. It is a requirement to have operator maintenance systemised as this provides: • Clear visibility and transparency for all maintenance work • Align to the requirements of the Glencore Cu Asset Management Framework • Promotes a joint ownership culture. For mobile plant pre-start check can be managed through a paper based or Fleet Management system.
3.3 Failure Identification	Operations shall be able to identify equipment that has entered into functional failure through condition monitoring using the human senses and diagnostic and monitoring systems. All defects shall be logged in the CMMS and E08 Maintenance Work Management followed.

Copyright © Glencore 2022

3.4 Housekeeping	Develop and Implement an initiative for operational housekeeping, an example would be 5S, see below: 5S is a Lean Manufacturing approach centred around 5 key steps: • Sort - All items, equipment, and work materials should be neatly arranged, and all unnecessary objects should be removed. The objective is to reduce clutter and make it easier to locate the resources needed for work. • Set in Order - All items, equipment, and work materials should be in optimal locations. The objective is to maximize accessibility, free up space, and prevent accidents from occurring by removing unnecessary obstacles. • Shine - The workspace, including all tools, equipment, and machinery, should be cleaned on a regular basis. The objective is to make the workspace safe, waste-free, and conducive to productivity. • Standardise - The processes for sorting, order, and cleanliness should be standardized and implemented across all offices and branches of operation. The objective is for all aspects and branches of operation to consistently gain the benefits of practicing sort, set in order and shine. • Sustain - Continuously and consistently practice the 5S methodology. The objective is to maximise the business's potential by removing all obstacles to productivity that is within the operation's control
3.5 Operational Variance Analysis and Response Plans	An Operational Variance and Response Plans shall be developed to provide operational performance requirements and Response actions should operational variance be outside performance requirements. Trigger Action and Response Plans are an example that will fulfil this requirement can be used to demonstrate the relationship between conditions in: • a normal environment • an abnormal environment (. e.g., equipment malfunctioning or overheating) • an emergency situation. • production rates The Plans should be: • be simple and robust. • be adequately resourced in terms of personnel and equipment, including monitoring systems. • focus on prevention and control through early detection. • set triggers through detailed knowledge of what is normal. • be regularly reviewed and revised as necessary. • be based on the best available on-site and off-site advice. An example TARP is provided at Appendix A
3.6 Monitor and Audit	Operator routines and practises shall be monitored on a regular basis, improvement actions identified and implemented
4.0 Skills Matrix Implemented	**Develop and implement a detailed skills matrix and training plan to underpin improving asset performance and safety. Refer to E2 Personnel Management**
4.1 Identify Operator Skills	Conduct Training Needs Analysis (TNA) to determine operator skill requirements.
4.2 Identify Maintainer Skills	Conduct Training Needs Analysis (TNA) to determine maintainer skill requirements.

Copyright © Glencore 2022

4.3 Measure Current Competency	Develop skills matrix for operator and maintainer and measure current skills against the skills matrix
4.4 Develop Training Program	Develop operator and maintainer training program using skills matrix as the basis for the training needs.
4.5 Implement Training Program	Develop multicriteria training programmes to enable training delivery through face to face, online, self-learning and on the job etc.
5.0 Integrated Reliability Approach	**Develop and implement an integrated approach to reliability improvements that incorporates operators, maintainers and technical functions**
5.1 Routine Review	• Asset performance against expected performance regarding risk, cost and production • Conduct Annual review of Operator standard checks and Response Plans developed in 3.5. • Conduct annual review of operator, maintainer and technical function skills
5.2 Identify Opportunities	Identify Reliability Improvement opportunities through collaborative teams to: • Understand the key causes of unreliability, these can be operational, maintenance, supply chain, Engineering and contractor management. • Prioritise initiatives and build business case. • Implement workable solutions. Allocate responsibilities to team members to develop project plans, seek approval, and implement projects
5.3 Review Previous Actions	Review the actions allocated to team members to ensure project delivery on time and on budget. Analyse the project benefits to see if any gaps exist regarding expected benefits
6.0 KPI's Developed and Measured	**Developing and implementing a series of operator KPIs (and their supporting metrics) to monitor asset performance improvement**
6.1 Develop and Implement Operator KPI's	Measure the performance of the assets and identify instances where operation occurs outside of the agreed plans or equipment operating envelopes. Any variance should be discussed with Operators so that explanations and corrective actions can be identified. • Identify what needs to be measured. ○ Understand what needs to be improved and why. KPI's should include factors that impact business outcomes including production and asset health. • Set targets. ○ Setting targets should be completed collaboratively with operators to ensure ownership. ○ Targets may need to be reset frequently to ensure that they are realistic and representative of operating context. (e.g., local throughput or recovery targets may need to be set daily based on forecast grades and production plans). • Collect information.

Copyright © Glencore 2022

	○ Data can be collected through manual collection, automated processes or integrated data analysis.
6.2 Equipment Score Cards	An equipment scorecard is a report that is provided to equipment operators. This same information is valuable to Managers/Superintendents/Supervisors and operator Trainers, to allow focused improvement for all operators and operating practices. A scorecard should: • Compare equipment operation practises between crews. • Trends to indicate process stability / improvement feedback. • Highlight equipment operation that require improvement. • Provide monthly scorecard feedback – minimum. • Provided immediate/shift feedback – ideal. Consideration for Operator Score Cards should be given to provide a vehicle by which consistent operation of equipment can be achieved. An example operator score card can be seen at Appendix C. The intent of the score cards to ensure that the business is in control of the asset and not the asset in control of the business.
6.6 Measure and Improve	Review Performance • Reviewing performance should be conducted as frequently as required to drive changes in Operational behaviours. • A collaborative approach should be taken to reviewing performance against targets that have been set. Where variance exists, it is important to identify and record the reasons, including when these reasons are outside the operator's control. Corrective actions should be identified and executed where there is a potential for the variance to continue or reoccur.

Copyright © Glencore 2022

4.3 Process RACI Description

The RACI matrix below in Table 1 addresses the high-level functional elements of the Operate for Reliability process. For the overall Asset Management RACI matrix, refer to the Asset Management Intranet page.

Site roles may differ to the roles identified in this RACI, which should be used as a guide for the development of site specific RACI, while maintaining the intent and content of the Manual.

Note: The HSE and Commercial Departments do not appear in this high-level RACI, although their involvement will be required from time to time dependent on the nature of the variance.

The codes used in the RACI analysis are defined as follows:

TABLE 4-3.1 PROCESS RACI

R	Responsible	The "doer" – the individual(s) who will perform an activity or directs others to perform the activity.
A	Accountable	"The buck stops here" –the individual who is ultimately "owns" the activity – there should only be one person
C	Consulted	person that needs to provide feedback and contribute to the activity
I	Inform	The individual(s) who needs to be informed after a decision or action is taken if relevant. (via 1-way communication)

Functional Role / Process Step	Operations Manager /Superintend	Operations Supervisor	Operator	Maintenance Manager/ Superintende	Maintenance Supervisor	Tradesperson	Reliability Engineer	Technical Services	Training
1.1 Responsibilities	A	R	I	C	I	I	C		
1.2 Interfaces between functions	A	I	I	R	C	C	C	C	
1.3 Equipment Health	A	C	C	R	C	I	R	C	
1.4 Asset Ownership	A	R	C	R	C	C	C	C	
2.1 Deploy operating specification	A	R	C	C	C	I	C	C	
2.2 Deploy maintenance specification	C	C	I	A	C	C	R	C	
2.3 Operate for Reliability Training	A	C	I	C	C	I	C	C	R
3.1 FMECA Implemented	C	C	C	R	C	C	R	C	
3.2 Standard operating practices	A	R	C	C	C	I	C	C	
3.3 Failures identified	A	R	C	C	C	I	C	I	
3.4 Housekeeping	A	R	R	R	R	R	R	R	
3.5 Operational Variance Analysis and Response Plans	A	R	C	C	C	I	C	R	
3.6 Monitor and Audit	A	R	C	C	C	I	C	C	
4.1 Identify operator skills	A	R	I	C	C	I	C	I	C
4.2 Identify maintainer skills	C	I	I	A	R	I	C	C	C
4.3 Measure current competency	A	R	C	C	I	I	C	C	C
4.4 Develop Training program	A	C	I	C	C	I	C	C	R

Copyright © Glencore 2022

Process Step \ Functional Role	Operations Manager /Superintend	Operations Supervisor	Operator	Maintenance Manager/ Superintende	Maintenance Supervisor	Tradesperson	Reliability Engineer	Technical Services	Training
4.5 Implement Training program	A	I	I	C	I	I	C	I	R
5.1 Routine Review	A	R	I	C	I	I	C	C	
5.2 Identify opportunities	A	R	C	R	C	C	C	C	
5.3 Review Previous actions	A	R	C	R	C	C	R	C	
6.1 Develop and Implement Operator KPI's	A	R	I	C	I	I	C	C	
6.2 Equipment Score Cards	A	R	C	R	C	I	C	C	
6.3 Measure and Improve	A	R	C	R	C	C	C	C	

TABLE 4-3.2 FUNCTIONAL DESCRIPTIONS

Function	Description
Operations Manager/Superintendent	Provides leadership in the short to medium term time horizon. Accountable for ensuring operators are competent, capable and that all equipment is operated within designated operating windows. Accountable for ensuring operator routines are developed and deployed and that equipment failure modes are identified before function system/equipment failure occurs.
Maintenance Manager/Superintendent	Provides leadership in the short to medium term time horizon. Responsible for establishing the maintenance plan, budget and managing the execution of maintenance activities. Ensure that the maintenance strategy is fit for purpose and participates in the development of the strategy and tactics.
Operations Supervisor	Responsible for ensuring that assets are operated within design limits and are done so safely. Provides input into the maintenance strategy development process as required. Responsible for ensuring operator routines are developed and deployed and that equipment failure modes are identified before function system/equipment failure occurs.
Operator	Responsible for ensuring that assets are operated within design limits and are done so safely. Provides input into the maintenance strategy development process as required. Provides input into the deployed of Operate for Reliability process and is consulted on certain aspects of the process.
Maintenance Supervisor	Provides input into the Operate for Reliability process deployment on a consulted basis. Ensures that all maintenance tasks are executed safely and to the desired quality, first time every time
Maintenance Trades	Responsible for carrying out planned and non-planned maintenance activities. Assists in the development of the maintenance tactics and provides input into the Operate for Reliability process

Copyright © Glencore 2022

Reliability Engineer	Responsible for the deployment of the corporate Equipment Operating and Maintenance Strategy Manual and the development and deployment of the site-specific Reliability processes and procedures.
Technical Services – Metallurgists, Process Engineers and Mining	Provide input and consultation into the Operate for Reliability process deployment. Responsible for the development and deployment of some Operational Variance Analysis and Response Plans
Training	Responsible for the development and deployment of training programmes and material in consultation with operations, maintenance and technical services.

5. Deployment

The deployment of the Operate for Reliability approach across all business areas is through:

- Awareness presentation
- Position description alignment
- Competency assessment
- Training, coaching, and mentoring

5.1 Awareness Presentation

This Manual is supported by a slide presentation and video to help facilitate communication and to raise awareness on how it contributes to the Asset Management Policy and business success.

5.2 Position Description Alignment

To effectively deploy this Manual, the essential process steps and activities need to be diligently mapped and assigned to approved positions within Glencore Cu.

The RACI matrix assists in the visualisation that everyone is working towards the same goal. It clarifies, not only the individual's responsibilities, but also that of others. It has proven to be an effective tool to quickly understand and close the gaps between current and desired practices and business process activities.

Position descriptions including KPIs need to be aligned to the RACI matrix and measures of success. This is the basis for ongoing performance reviews and the creation and execution of individual development plans.

5.3 Competency Assessment

This Manual is supported by an Asset Management competencies assessment process and each site is responsible for assessing Asset Management competencies which is used to:

- Guide recruitment, selection, development, and deployment of people in asset management roles across the company
- Provide employees responsible for managing assets with feedback, development, coaching and recognition for their performance.
- Map, build and manage the asset management talent pool across the company.
- Identify strengths and development needs in asset management practitioners.

Copyright © Glencore 2022

5.4 Training, Coaching and Mentoring

Coaching and mentoring is another way to assist employees improve their understanding of their role and to improve performance.

Coaching is generally about improving performance in the short term where mentoring is about achieving high performance and career aspirations over a longer term.

Coaching or mentoring requires addressing that a person is not currently up to the required level to perform their role as effectively and efficiently as they could. The build-up of skills to perform efficiently and effectively requires a large investment of time and cooperation.

Coaches and Mentors need to be carefully selected and must be committed for the long term to effect maximum success.

6. Measures of Success

TABLE 6-1 MEASURE OF SUCCESS

Metric	Operator Routines Compliance
Definition	**Operator Routines –** Work undertaken by operations personnel to ensure that the operating equipment is performing as expected and to identify issues and to create work requests to rectify the issues.
Performance Indicator	95% of Operator Routines complete
Calculation	Number of Operator Routine complete/ Number of Operator Routines Planned x 100
Example	PM02's Work Orders in SAP for Operations work centres
Rational	Operations Personnel play a vital role in the identification of performance issues and Operator Routines provide a structured and formal means by which these issues can be identified and addressed

Metric	Equipment Scorecards
Definition	Equipment Scorecards are a means by which the performance of the operational equipment can be measured, monitored, compared and improved.
Performance Indicator	95% of Critical Equipment contained within an Equipment Scorecard
Calculation	Total number of Critical Equipment covered by Equipment Score Card/Total number of Critical Equipment
Example	NA
Rationale	Operating critical equipment consistently within operational parameters has a positive impact on the amount and quality of product that is produced. Measuring, monitoring, comparing and improving the way in which critical equipment is operated enables business objectives to be achieved within desired timeframes and within expected costs.

Copyright © Glencore 2022

Metric	% Critical System with Operational Variance Analysis and Response Plans
Definition	Percentage of Critical System that have plans in place to enable rapid response to operational variance.
Performance Indicator	100% of Critical System to have Plans in place.
Calculation	*N/A*
Examples	Trigger Action and Response Plan (TARP) Mobile Plant variance to follow Maintenance Work Management process as defined in Element 08
Rationale	An Operational Variance Analysis and Response Plan provides a predefined structured means by which variances in operational parameters can be identified and corrective actions put in place to eliminate or reduce the significance of the variable. For

Metric	Production Cost Per Tonne
Definition	The total cost of producing one tonne of product
Performance Indicator	Recorded monthly and trending down on an annual basis.
Calculation	*N/A*
Rationale	As Operate for Reliability is embedded into the business there are a number of value-add benefits to be realised from process compliance, these include but not limited to: • Reduced operational downtime attributable to breakdowns. • Increase in equipment life due to operating equipment within operating parameters. • Reduced energy cost. • Reduced maintenance cost. • Improved safety performance. • Reduced risk due to operational consistency and reduction in operational variation These benefits when achieved will reduce the cost per tonne of product produced

Metric	Effective Utilisation
Definition	Percentage of Calendar Time an asset or plant area was performing its primary function
Performance Indicator	95% of time performing primary function when available.
Calculation	*Operating time/Calendar time*
Rationale	Derived from Glencore Cu Time Usage Model and Global KPI's

Copyright © Glencore 2022

7. Improvement

The process to review and continually improve the approach and deployment for Operate for Reliability is scheduled annually or when there is a sudden change in the Business and Asset Management objectives. As the organization matures a longer review period could be considered.

8. References

The following documents provide detailed support material that should be read in conjunction with this Manual to provide a comprehensive understanding of Operate for Reliability.

TABLE 8-1 REFERENCES

DOCUMENT TITLE	DESCRIPTION
Asset Management Framework	This document defines the overall direction of asset management across Glencore Copper.
Guideline for Operate for Reliability	This document defines previous Operate for Reliability Guideline within the Glencore business
Glencore Cu Time Usage Model	The current time usage model within Glenore Cu
Glencore Cu Global KPI's	The current suite of Global KPI's within Glencore CU

9. Revision History

VERSION NO.	REASON FOR REVISION	DATE
0.1	New Manual to support the Asset Management Policy	26/08/2020
0.2	Gather Documents and Create Draft Manual	29/01/2021
0.3	Review and Signoff 1st Draft Manual with Sites	03/02/2021
0.4	Reformat and Issue for Final Review	23/02/2021
1.0	Update Site Signoff and Publish	11/10/2021

Copyright © Glencore 2022

10. Appendix A – TARP Example

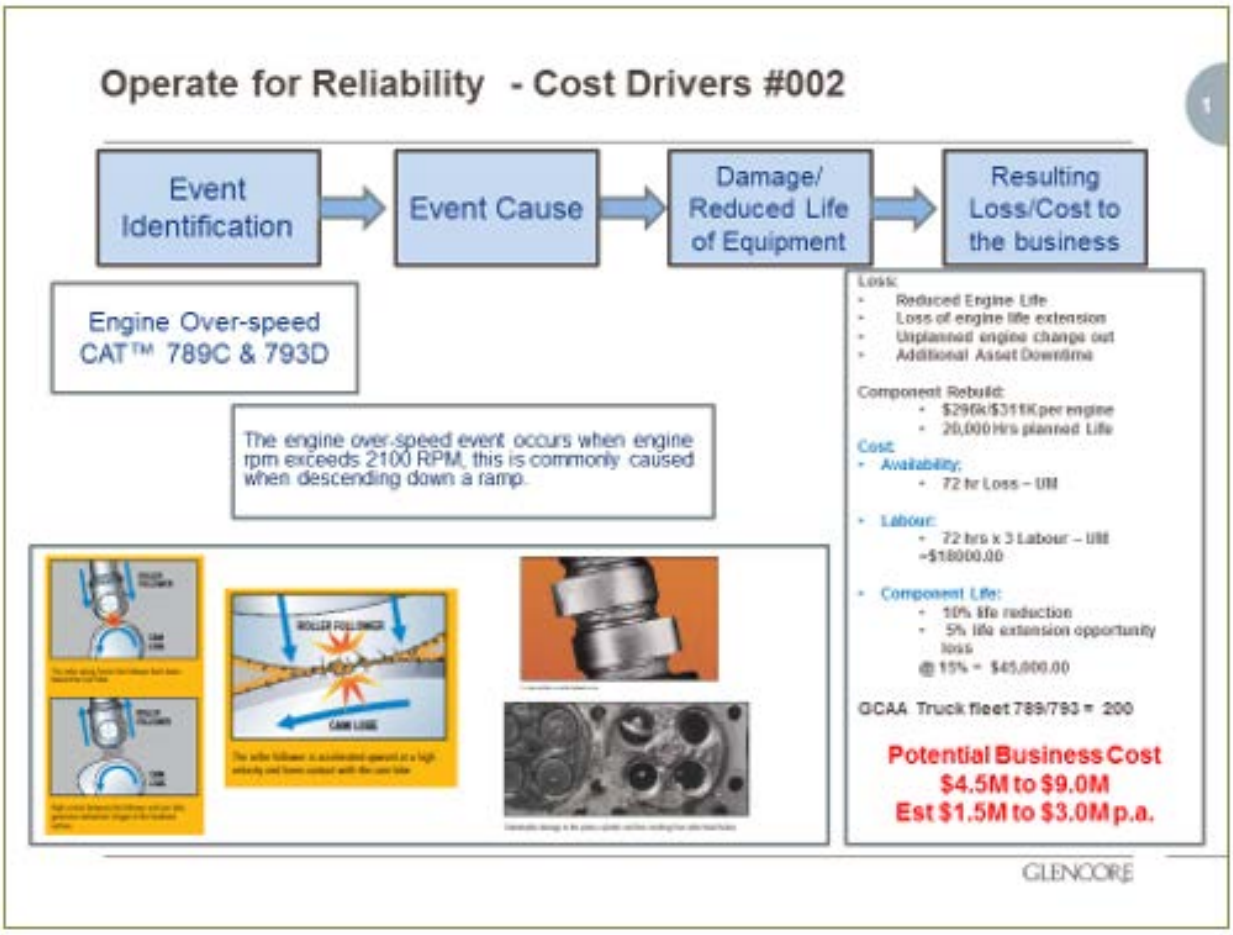
Situation: Low rougher recovery based on MSA hourly averages.

	Normal - Rougher recovery at or above target	Rougher recovery up to 1% below target for 1 hour	Rougher recovery up to 1% below target for 2 hours or up to2% below target	Rougher recovery up to 1% below target for 4 hours or above 2% below target.
Flotation operator	o Perform routine plant checks as per the checklist o Record plant set points on the log sheet o Maintain an accurate OIS record o Attend and contribute to 3W meeting 3 times per shift o Ensure reason for operating outside the range set on the plant monitoring sheet are logged and on the 3w sheets o Review current MSA results and trends to control flotation performance, may be trending up or down o As soon as they are available review the last shift results from the Lab	o Review any changes to the process and determine if they may have had a detrimental effect on performance o Check all MSA streams are delivering flow to the MSA and that the MSA dipping all zones o Visually check rougher, bank levels, reagent additions and levels are correct o Review the plant monitoring sheet for out of range settings. o Review fish bone analysis from previous events	o Co operate with the plant metallurgist and under take suggested changes o Continue to perform routine checks as per the checklist	o Co operate with the plant metallurgist and under take suggested changes o Continue to perform routine checks as per the checklist
Control room operator	o Using the DCS and Pi monitor flotation performance and communicate results to the flotation operator o Communicate tonnage and feed grade changes to the Flotation operator	o Using the DCS and Pi monitor flotation performance and communicate results to the flotation operator o Communicate tonnage and feed grade changes to the Flotation operator	o Using the DCS and Pi monitor flotation performance and communicate results to the flotation operator o Communicate tonnage and feed grade changes to the Flotation operator o Have the crusher, grinding and dewatering areas checked for possible causes of contamination (oils or chemicals etc.)	o Using the DCS and Pi monitor flotation performance and communicate results to the flotation operator o Communicate tonnage and feed grade changes to the Flotation operator
Production supervisor	o Ensure plant checks are done o Attend 3W meeting 3 times per shift	o Ensure plant checks are done o Attend 3W meeting 3 times per shift	o Ensure plant checks are done o With the flotation operator check the rougher for possible issues o Attend 3W meeting 3 times per shift o Notify the plant metallurgist and the production superintendent	o Facilitate an extra 3W meeting to record agreed actions going forward o Ensure actions from the extra 3W meeting are carried out
Plant metallurgist	o Monitor and communicate opportunities for improvement o Attend 3W meeting 3 times per shift	o Monitor and communicate opportunities for improvement o Attend 3W meeting 3 times per shift o Review lab versus MSA accuracy and communicate o Visually inspect the rougher performance and suggest possible changes	o Review steps taken so far and if there is any thing else that can be done o Notify the metallurgical superintendent.	o Attend the extra 3W meeting
Production superintendent			o Review steps taken so far and if there is any thing else	o Ensure all available resources are actively working towards a resolution o Attend extra 3W meeting
Metallurgical superintendent			o Review steps taken so far and if there is any thing else	o Ensure metallurgical sampling regime is carried out o Attend extra 3W meeting
Maintenance team	o Perform repairs or checks upon request	o Perform repairs or checks upon request	o Perform repairs or checks upon request	o Perform repairs or checks upon request

Copyright © Glencore 2022

11. Appendix B -Operator Scorecard Example

Engine Overspeed – Cost Impact

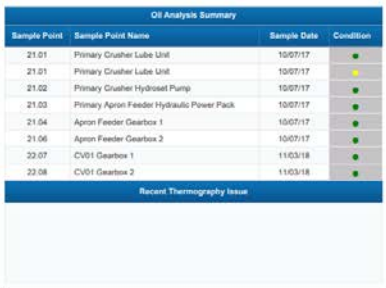


Copyright © Glencore 2022

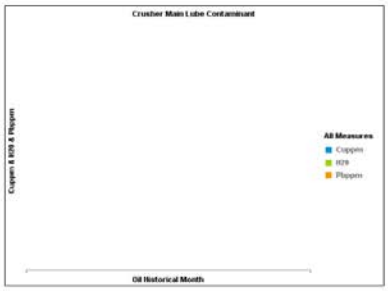
12. Appendix C – Equipment Score Card Example

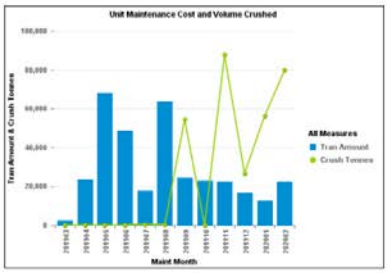
Crusher and Reclaim Summary











Asset Management, Glencore Copper
11 October 2021, Version 1.0

Contributors:

- Maximo Rojas (PASAR)
- Felipe Bunout (Lomas Bayas)
- Ryan Turner (Mutanda Mining)
- Danai Saotayanan (Queensland Metals - Processing and Smelting)
- Tom Barber (Ernest Henry Mining)
- Mark Backman (Queensland Metals – Mining)
- Christopher Dwan (Queensland Metals – Processing and Smelting)
- Neale Laffin (Queensland Metals – Processing and Smelting)
- Felipe Moraga (Technology Transformation and Operational Excellence South America
- Osvaldo Zamorano (Operational Excellence Metallurgical Assets),
- Peter Zuliani (Operational Excellence Metallurgical Assets),
- Dino Moll (Glencore Copper Asset Management - Australia),
- Rohan Reisener (Glencore Copper Asset Management - Australia).

Supported by the AM Network Pty Ltd

Copyright © Glencore 2022

13. Appendix D – Stakeholder Signoff

Stakeholder Name	Site or Department	Contact	Stakeholder Signoff
Maximo Rojas	PASAR	Maximo.Rojas@pasar.com.ph	DocuSigned by:
Felipe Bunout	Lomas Bayas	Felipe.Bunout@glencore.cl	DocuSigned by: Felipe Bunout
Ryan Turner	Mutanda Mining	Ryan.Turner@mumisprl.com	DocuSigned by:
Christopher Dwan	Queensland Metals – Processing and Smelting	Christopher.Dwan@glencore.com.au	DocuSigned by: Chris Dwan C172BE85E4FD43F
Neale Laffin	Queensland Metals – Processing and Smelting	neale.laffin@glencore.com.au	DocuSigned by: Neale Laffin EF5E1F712BF84FB
Tony Archer	AM Network	tarcher@amnetwork.com.au	DocuSigned by: Tony Archer
Felipe Moraga	Technology Transformation and Operational Excellence South America	Felipe.Moraga@glencore.cl	DocuSigned by: Felipe Moraga C0B0EA8208F94ED
Osvaldo Zamorano	Operational Excellence Metallurgical Assets	Osvaldo.Zamorano@glencore.cl	DocuSigned by: Osvaldo Zamorano 1C2AAF86BFA3471
Peter Zuliani	Operational Excellence Metallurgical Assets	Peter.Zuliani@glencore.ca	DocuSigned by: Peter Zuliani 7BD3EC89C3C846C
Dino Moll	Asset Management - Australia	Dino.Moll@glencore.com.au	DocuSigned by: Dino Moll 413A4F0615AF4ED
Rohan Reisener	Asset Management - Australia	rohan.reisener@glencore.com.au	DocuSigned by: Rohan Reisener 9B7E3453600948D

Disclaimer

Copyright © Glencore 2022

This document is the property of Glencore and contains information that is the Intellectual Property of Glencore or is otherwise proprietary to Glencore. This document and the information contained herein shall not be made available to any person or persons not authorised to receive it nor shall it be reproduced, copied, loaned, exhibited or used for any other purpose whatsoever, in whole or in part, without the prior written consent of Glencore.

Copyright © Glencore 2022

Notes

Copyright © Glencore 2022

Notes

Copyright © Glencore 2022


GLENCORE
07

07

Operate for Reliability

Rohan Reisener
Manager – Asset Management Strategy and Development
Glencore Copper Operational Excellence
T: +61 7 3833 8408 M: +61 478 316 195
rohan.reisener@glencore.com.au
160 Ann St, Brisbane QLD 4000